Monetary Policy Communication under Learning and Strategic Motives

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February 25, 2026

Working Paper — Preliminary Draft

Abstract

This paper examines how central-bank communication affects macroeconomic stability when expectations are formed under constant-gain learning and strategic information acquisition. Agents combine a private estimate with a public policy signal, and equilibrium signal weights reflect incentives to coordinate or differentiate forecasts. Belief uncertainty enters one-step-ahead forecasts through a covariance correction and feeds back into equilibrium outcomes. A central implication is that E-stability need not require strong policy inertia: strategic signal extraction can attenuate the destabilizing impact of belief revisions even when interest-rate smoothing is only moderate. We show that transparency anchors higher-order beliefs and weakens the covariance-feedback amplification of monetary shocks, with the strongest effects under coordination motives. Quantitatively, relative to a rational-expectations benchmark, the combined effect of strategic information acquisition and the covariance channel increases output-gap IRF fluctuations by about 0.3% in our calibration. Overall, the results suggest that effective communication design must account not only for signal precision, but also for how strategic forecasting incentives reshape information weights and the belief-uncertainty feedback that governs learnability and stability.

1 Introduction

Motivation A growing empirical literature studies how the "tone" of speeches of monetary policy communication moves asset prices and market liquidity (Swanson and Jayawickrema, 2023; Granziera et al., 2025) In preliminary work, we replicate the semantic-orientation measure in Tobback et al. (2017) to quantify the "tone" of Fed speeches and

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relate it to changes in the average EUR/USD spot bid–ask spread. We then compare this to an ambiguity index[1] that we construct as a proxy for the precision of policy communication. Two patterns stand out in Figures 1 and 2. First, spread changes are more strongly related to our ambiguity measure than to tone. Second, the market reacts sharply even to speeches delivered during the FOMC blackout window, when these speeches should be uninformative. Together, these facts suggest that what matters for expectations and market outcomes is not only *what* the central bank says, but also *how precisely* its communication anchors beliefs. This motivates the central question of the paper: what role does the precision of public announcements play in expectation formation and, ultimately, in macroeconomic stability?

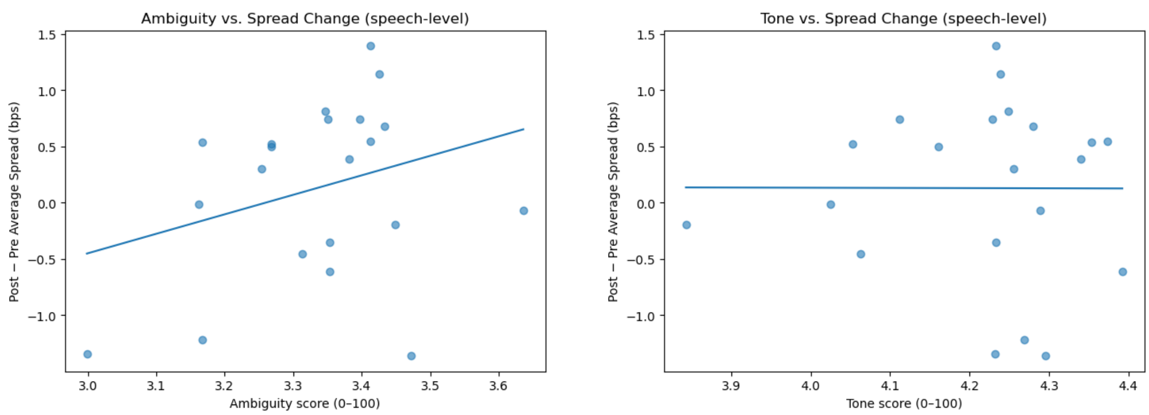


Figure 1: Speech-level EUR/USD average bid–ask spread changes around Fed speeches.

Two strands of theory approach this question using different primitives. In learning-based models with central-bank informational advantage, transparency enforces stability by reducing perceived uncertainty and facilitating belief updating (Evans and Honkapohja, 2001; Eusepi, 2005; Eusepi et al., 2005). However, this literature typically relies on decreasing-gain learning and therefore abstracts from an additional source of short-run fluctuations induced by persistent belief uncertainty: when beliefs remain imprecise, the economy becomes endogenously noisier and learning itself can be impaired. Importantly, standard learning models rarely incorporate strategic information acquisition. Recent evidence suggests that forecasters overweight idiosyncratic signals and underweight common information (including policy announcements) in order to differentiate from the crowd (Adam et al., 2025; Gemmi and Valchev, 2026). In contrast, the information-design literature emphasizes that greater public-signal precision can be socially harmful in coor-

[1] We implement a hedging-intensity measure, defined as the frequency of hedging or conditional language (e.g., "may/might/could," "depends," "uncertain," "monitoring," "risks are"), normalized by speech length (per sentence or per 1,000 words).

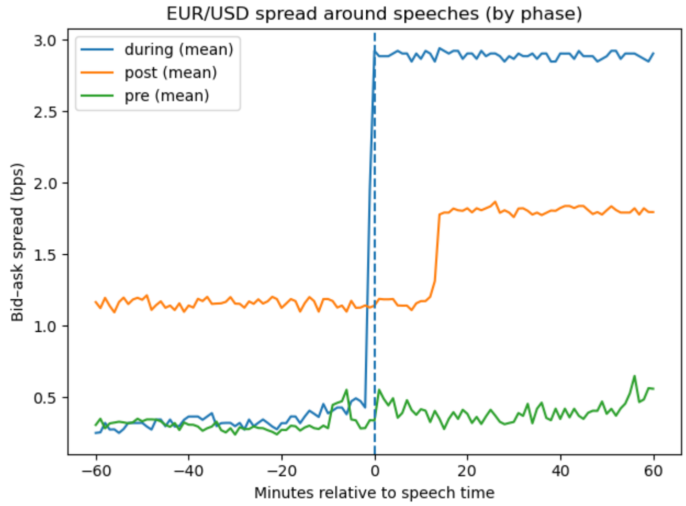


Figure 2: Average EUR/USD bid–ask spread in event time around Fed speeches delivered before, during and after the "Black-out" period . The series report the mean spread in the pre-speech, during-speech, and post-speech windows, with the vertical dashed line indicating the speech start ($t = 0$).

dination environments because it induces over-reliance on the public signal (Morris and Shin, 2002; Hellwig and Veldkamp, 2009). Motivated by these considerations, this paper studies how policy-announcement precision affects learning and stability when agents extract public and private signals under strategic motives.

Main Findings Our contribution is to combine learning and strategic signal extraction in a New Keynesian model in which agents estimate forecasting coefficients via constant-gain learning, and policy transparency is modeled as the relative precision of the public signal compared with private signal. The model delivers three main findings. First, communication precision matters for learnability because belief uncertainty enters one-step-ahead forecasts through a covariance correction, which feeds directly into the actual law of motion and amplifies forecast errors and belief revisions. Second, stronger policy inertia improves learnability by reducing endogenous variation in the environment that drives forecast errors and belief updating. Third, the stabilizing role of precision depends on strategic incentives: under coordination motives, agents place greater equilibrium weight on public information, so deteriorations in public-signal quality generate larger co-movement in beliefs and outcomes; whereas under substitution motives, the same change in precision has weaker aggregate effects.

Related literature This paper connects to two strands of research on the effects of monetary policy communication. First, it relates to the learning literature that studies determinacy, learnability, and stability under subjective expectations. Our baseline framework builds on Evans and Honkapohja (2003), Eusepi (2005), Eusepi (2007), Bullard and Mitra (2007), and Bullard and Suda (2016), but differs in a key aspect: agents use constant-gain learning and form one-step ahead forecasts in an environment where belief uncertainty matters. In particular, belief uncertainty enters into forecasts through a covariance correction, which feeds back into equilibrium outcomes and can generate short-run fluctuations even when long-run learnability conditions hold.

Second, our paper relates to the information and coordination literature in which agents strategically weight public and private signals under rational expectations. Classic contributions such as Morris and Shin (2002), Walsh (2007), Gosselin et al. (2007), and Hellwig and Veldkamp (2009) emphasize that more precise public signals can be welfare-reducing when they induce excessive coordination on a potentially noisy common signal. We depart from the rational-expectations benchmark by studying how communication precision operates in a learning environment where agents face strategic forecasting incentives. This perspective clarifies how equilibrium information weights and learnability interact, and when transparency stabilizes (or destabilizes) the economy.

Finally, our framework provides a learning-based foundation for recent empirical evidence on strategic disagreement and biased professional forecasts Adam et al., 2025; Gemmi and Valchev, 2026. It also complements the growing empirical literature on the market effects of central bank communication and "tone" Tobback et al., 2017; Swanson and Jayawickrema, 2023; Granziera et al., 2025 by highlighting that what matters is not only the direction of communication, but also its precision: clarity affects how agents weight information, update beliefs, and ultimately the stability of macroeconomic dynamics.

This paper is organized as follows. Section 2 presents the baseline model and derives the determinacy and E-stability conditions. Section 3 extends the baseline by introducing a forecast-competition game in the first stage of each period. Section 4 describes the calibration and discusses impulse responses to monetary policy shocks. Section 5 concludes.

2 Baseline Model

2.1 Environment

Households & Firms. Following Eusepi (2005), we adopt a standard New Keynesian model as in Woodford (2003). A continuum of households allocates consumption across differentiated goods and supplies labor, while monopolistically competitive firms produce using labor and face quadratic price adjustment costs. The log-linearized equilibrium can be written as a forward-looking IS curve,

$$x_t = \mathbf{E}^*_{t-1} x_{t+1} - \sigma^{-1} \mathbf{E}^*_{t-1} (i_t - \pi_{t+1} - r^n_t)\,, \tag{1}$$

where x_t denotes the output gap, and r^n_t is the (deviation) from natural real rate $(\beta^{-1} - 1)$.

Firms' price-setting decisions imply a Phillips curve,

$$\pi_t = \beta \mathbf{E}^*_{t-1} \pi_{t+1} + \xi\, \mathbf{E}^*_{t-1} (\kappa x_t + \varsigma i_t) + u_t, \tag{2}$$

where u_t is a cost-push shock. $\kappa = \frac{\theta+\chi}{1-\theta} + \sigma$, with θ denoting the output elasticity of labor, χ the consumption-curvature parameter, and σ the inverse intertemporal elasticity of substitution.. The liquidity term is $\varsigma = \frac{\beta\nu}{1+(1-\beta)\nu}$, where ν governs the strength of the cash-in-advance constraint. As in Eusepi (2005), decisions are made before (x_t, π_t) are observed, but note that those expectations are subjective conditional on the information set available at $t-1$.

Forecast-based monetary policy rule. Monetary policy is characterized by a forward-looking Taylor-type rule with interest-rate smoothing,

$$i_t = \delta_\pi \mathbf{E}^*_{t-1} \pi_{t+1} + \delta_x \mathbf{E}^*_{t-1} x_{t+1} + \delta_L i_{t-1} + \epsilon_t, \tag{3}$$

where ϵ_t is a monetary policy shock (or control error). The functional form is assumed to be common knowledge, but agents are uncertain about the coefficient vector $\delta \equiv (\delta_\pi, \delta_x, \delta_L)'$. Agents therefore treat φ as an additional latent object and update beliefs using the observed policy rate i_t and the associated regressors $f_t \equiv \left(\mathbf{E}^*_{t-1}\pi_{t+1},\, \mathbf{E}^*_{t-1}x_{t+1},\, i_{t-1}\right)'$. Following Bullard and Mitra (2007), we treat (3) as an operational class of policy rules and focus on the implications for determinacy and E-stability rather than deriving (3) from an explicit policy-loss problem. Consistent with Bullard and Mitra (2007), Figure 3 illustrates that an active inflation response and sufficiently strong policy inertia are key for determinacy in forecast-based rules. Due to space constraints, the corresponding determinacy analysis is delegated to Appendix A.1.

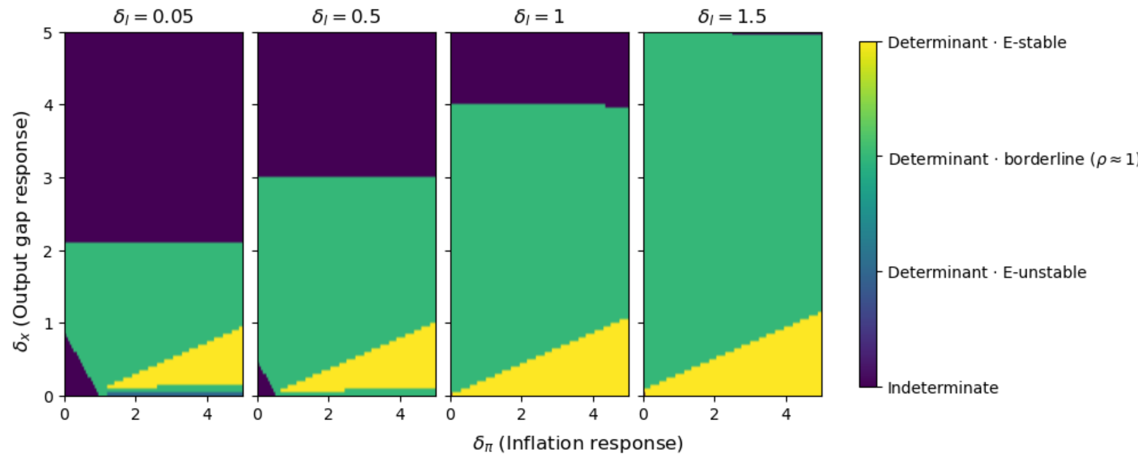


Figure 3: Determinacy and E-stability regions in the baseline model across policy feedback coefficients (δ_π, δ_x) for different degrees of interest-rate smoothing δ_ℓ. Colors indicate whether the RE equilibrium is indeterminate, determinate but E-unstable, borderline E-stable ($\rho \approx 1$), or (strongly) E-stable.

2.2 Rational Expectation Equilibrium and E-stability Conditions

Minimal State Variables Solution Let the endogenous state be $V_t = [x_t, \pi_t, i_t]'$. Stacking (1), (2) and (3) yields

$$V_t = \underbrace{\begin{bmatrix} 0 & 0 & -\sigma^{-1} \\ \xi\kappa & 0 & \xi\varsigma \\ 0 & 0 & 0 \end{bmatrix}}_{\Omega_1} \mathbf{E}_{t-1} V_t + \underbrace{\begin{bmatrix} 1 & \sigma^{-1} & 0 \\ 0 & \beta & 0 \\ \delta_x & \delta_\pi & 0 \end{bmatrix}}_{\Omega_2} \mathbf{E}_{t-1} V_{t+1}$$
$$+ \underbrace{\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \delta_L \end{bmatrix}}_{\Omega_3} V_{t-1} + \underbrace{\begin{bmatrix} \sigma^{-1} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\Omega_4} \begin{bmatrix} r_t^n \\ u_t \\ \epsilon_t \end{bmatrix}. \tag{4}$$

$X_t \equiv [r_t^n, u_t, \epsilon_t]'$ collects exogenous shocks, which follow a first-order VAR with diagonal persistence,

$$X_t = HX_{t-1} + \varepsilon_t^X, \qquad H = \mathrm{diag}(\rho_r, \rho_u, \rho_\epsilon), \quad \varepsilon_t^X \sim \mathcal{N}(0, \Sigma^X), \tag{5}$$

with ε_t^X independent of initial beliefs.

Under the standard minimal-state-variable (MSV) restriction, Appendix A.2 shows that any linear REE can be written as

$$V_t = A + BV_{t-1} + CX_t,$$

where (A, B, C) are constant matrices of conformable dimensions.

Perceived law of motion (PLM). Agents do *not* know the true coefficients that govern (4)–(5). Instead, they posit that the (vectorized) coefficient vector follows a time-varying parameter law,

$$\theta_t = \theta_{t-1} + \eta_t, \qquad \eta_t \sim \mathcal{N}(0, R_{1,t}), \qquad R_{1,t} \succeq 0, \tag{6}$$

where $R_{1,t}$ is the (possibly time-varying) parameter-drift covariance.

At the beginning of period t, agents observe (V_{t-1}, X_{t-1}) and form forecasts conditional on the information set $\mathcal{I}_{t-1} \equiv \{V_{t-1}, X_{t-1}\}$. After (V_t, X_t) are realized, they update beliefs using the resulting projection error.

Let

$$z_t = \begin{bmatrix} V_{t-1} \\ X_{t-1} \end{bmatrix} \in \mathbb{R}^6, \qquad \Phi_t = [\Theta_t^V \;\; \Theta_t^X] \in \mathbb{R}^{3\times 6},$$

and define $\theta_t = \text{vec}(\Theta_t) \in \mathbb{R}^{18}$ and $Z_t = z_t' \otimes I_3$. The perceived law of motion is

$$\begin{aligned} V_t &= \Theta_t z_t + e_t^V, \qquad e_t^V \sim \mathcal{N}(0, R_{2,t}) \\ &= Z_t \theta_t + e_t^V. \end{aligned} \tag{7}$$

For consistency, we work with the vectorized representation in what follows.

Kalman Filter & Recursive Least Square Learning Denote the conditional expected value of these parameters as $\hat{\theta}_{t|t-1} = \mathbf{E}[\theta_t|\mathcal{I}_{t-1}]$, and conditional covariance as $\hat{\Sigma}^{\theta}{}_{t|t-1} = \mathbf{E}[(\theta_t - \hat{\theta}_{t|t-1})(\theta_t - \hat{\theta}_{t|t-1})']$. With both (6) and (7), agents update their estimates by regressing the estimate errors on the forecast errors of the observable states so that:

$$\underbrace{\theta_t - \hat{\theta}_{t|t-1}}_{\text{Estimate errors}} = \beta_t \underbrace{\left(V_t - Z_t \theta_{t|t-1}\right)}_{\text{Forecast errors}} + \underbrace{\varsigma_t}_{\text{Random noise}} .$$

Assume that the regressors and random noise are independent, the conditional belief of latent parameters evolve according to the following recursive equations:

$$\hat{\theta}_{t|t} = \hat{\theta}_{t|t-1} + \beta_t \left(V_t - Z_t \hat{\theta}_{t|t-1} \right) \tag{8}$$

$$\hat{\Sigma}^{\theta}_{t|t} = \hat{\Sigma}^{\theta}_{t|t-1} - \beta_t \left(Z_t \, \hat{\Sigma}^{\theta}_{t|t-1} \, Z_t^{'} + R_{2,t} \right) \beta_t^{\prime} \tag{9}$$

$$\beta_t = \left(Z_t \, \hat{\Sigma}^{\theta}_{t|t-1} \, Z_t^{'} + R_{2,t} \right)^{-1} \hat{\Sigma}^{\theta}_{t|t-1} Z_t^{'}. \tag{10}$$

, where the conditional covariance $\Sigma^{\theta}_{t|t-1} = \Sigma^{\theta}_{t-1|t-1} + R_{1,t-1}$. Sargent (1999) and McCulloch (2025) show that if $R_{1,t}$ is proportional to $\hat{\Sigma}_{t-1|t-1}$ and $R_{2,t}$ is constant, then the recursive dynamic between Kalman Filter and recursive least squared learning is identical. Suppose that the constant learning gain is denoted as $g = 1 - \lambda$, where λ represents the forgetting / discounting factor on historical observation, then 8 and 9 become:

$$\hat{\theta}_{t|t} = \hat{\theta}_{t|t-1} + g(R_t)^{-1} Z_t^{'} (V_t - Z_t \hat{\theta}_{t|t-1}) \tag{11}$$

$$R_t = R_{t-1} + g(Z_t^{'} Z_t - R_{t-1}), \tag{12}$$

where $R_t = \hat{\Sigma}^{-1}_{t|t}$. Details of this correspondence can be found in the Appendix A.4. [2]

Given beliefs $(\hat{\theta}_{t|t-1}, \hat{\Sigma}^{\theta}_{t|t-1})$, agents form the period-t forecast

$$\mathbb{E}_{t-1}[V_t] \; = \; Z_t \, \hat{\theta}_{t|t-1}. \tag{13}$$

For the one-step-ahead forecast, note that $V_{t+1} = Z_{t+1}\theta_{t+1} + e^V_{t+1}$ under the PLM. Using $\mathbb{E}_{t-1}[e^V_{t+1}] = 0$, we obtain

$$\mathbb{E}_{t-1}[V_{t+1}] = \mathbb{E}_{t-1}[Z_{t+1}\theta_{t+1}] = \mathbb{E}_{t-1}[Z_{t+1}] \, \mathbb{E}_{t-1}[\theta_{t+1}] + \mathrm{Cov}_{t-1}(Z_{t+1}, \theta_{t+1}). \tag{14}$$

Under the random-walk belief (6), the predicted mean satisfies $\mathbb{E}_{t-1}[\theta_{t+1}] = \hat{\theta}_{t|t-1}$. Also, only the V_t-block in z_{t+1} is correlated with θ_{t+1}; the intercept and X_t are independent of θ. Using the measurement equation $V_t = Z_t\theta_t + e^V_t$ and assuming e^V_t is independent of θ_t

[2] An alternative approach to proceed to the E-stability analysis is to find the steady-state $\hat{\Sigma}_t$ by assuming $R_{1,t} = 0$ and R_2 is constant and solve the Discrete Time Algebraic Ricatti equation, which resembles the perpetual learning dynamic with a constant gain learning g.

conditional on $\mathcal{I}_{t-1}$, covariance identity can be restated as

$$\begin{aligned}\text{Cov}_{t-1}(V_t, \hat{\theta}_t^V) &= \mathbb{E}_{t-1}\Big[\big(V_t - Z_t\hat{\theta}_{t|t-1}\big)\big(\theta_t^V - \hat{\theta}_{t|t-1}^V\big)'\Big] \\ &= Z_t\,\hat{\Sigma}_{t|t-1}^{\theta}\,\text{vec}(J_V') \\ &= Z_t\left((1-g)\,R_{t-1}\right)^{-1}\text{vec}(J_V').\end{aligned}$$

where J_v identifies which components of θ_t (and hence which rows/columns of $\text{Var}(\theta_t)$) correspond to the coefficients on V_{t-1}. This term captures *belief uncertainty* because it measures how unexpected realizations of macro outcomes translate into revisions of the perceived law of motion. When beliefs are uncertain, a given macro surprise induces a larger revision in the coefficients on V_{t-1} (the block $\hat{\theta}_t^V$), strengthening the statistical comovement between V_t and belief revisions.

Combining terms, we obtain

$$\mathbb{E}_{t-1}[V_{t+1}] = \mathbb{E}_{t-1}[Z_{t+1}]\,\hat{\theta}_{t|t-1} + Z_t\left((1-g)\,R_{t-1}\right)^{-1}\text{vec}(J_V') \tag{15}$$

T-map & E-stability conditions Using $X_t = HX_{t-1} + \varepsilon_t^X$, we obtain the ALM as

$$\begin{aligned}V_t = {} & \Omega_1\left(Z_t\,\hat{\theta}_{t|t-1}\right) + \Omega_2\left(\mathbb{E}_{t-1}[Z_{t+1}]\,\hat{\theta}_{t|t-1} + Z_t\left((1-g)\,R_{t-1}\right)^{-1}\text{vec}(J_V')\right) \\ & + \Omega_3\,V_{t-1} + \Omega_4\,HX_{t-1} + \eta_t^V, \\ & + \left((\Omega_1 + \Omega_2\hat{\Theta}_{t|t-1}^V)\hat{\Theta}_{t|t-1}^V + \Omega_3\right)V_{t-1} + \Omega_2 Z_t\left((1-g)\,R_{t-1}\right)^{-1}\text{vec}(J_V') \\ & + \left((\Omega_1 + \Omega_2\hat{\Theta}_{t|t-1}^V)\hat{\Theta}_{t|t-1}^X + \Omega_2\hat{\Theta}_{t|t-1}^X H + \Omega_4 H\right)X_{t-1} + \eta_t^V\end{aligned} \tag{16}$$

where $\eta_t^V \equiv \Omega_4\,\varepsilon_t^X$. To ease the notation burden for the following analysis, the second line expresses the ALM in matrix with $\text{vec}(\Theta_{t|t-1}^V) = \theta_{t|t-1}^V$ and $\text{vec}(\Theta_{t|t-1}^X) = \theta_{t|t-1}^X$. So, we define the T-map as: $T(\hat{\Theta}_{t|t-1}; R_{t-1}) = [T^V(\hat{\Theta}_{t|t-1}; R_{t-1}),$
$T^X(\hat{\Theta}_{t|t-1}; R_{t-1})]$, where

$$T^V(\hat{\Theta}_{t|t-1}; R_{t-1}) = \Omega_1\hat{\Theta}_{t|t-1}^V + \Omega_2\big(\hat{\Theta}_{t|t-1}^V\hat{\Theta}_{t|t-1}^V + R_{t-1}^V\big) + \Omega_3, \tag{17}$$

$$T^X(\hat{\Theta}_{t|t-1}; R_{t-1}) = \Omega_1\hat{\Theta}_{t|t-1}^X + \Omega_2\big(\hat{\Theta}_{t|t-1}^V\hat{\Theta}_{t|t-1}^X + \hat{\Theta}_{t|t-1}^X H + R_{t-1}^X\big) + \Omega_4 H, \tag{18}$$

where $\big[\,R_{t-1}^V\;R_{t-1}^X\big] = \big((1-g)R_{t-1}\big)^{-1}\text{vec}(J_v')$ so that $Z_t\big((1-g)R_{t-1}\big)^{-1}\text{vec}(J_v') = R_{t-1}^V V_{t-1} + R_{t-1}^X X_{t-1}$.

With the above, we can represent (11) in the stochastic recursive form,

$$\hat{\theta}_{t|t} = \hat{\theta}_{t|t-1} + g(R_t)^{-1} Z_t' \Big[\Big(T(\hat{\Theta}_{t|t-1}; R_{t-1}) - \hat{\Theta}_{t|t-1} \Big) z_t + \eta_t^V \Big] \tag{19}$$

Proposition 1 (E-stability)**.** Let Θ^* be an MSV REE satisfying $\Theta^* = T(\Theta^*; \bar{S})$.The equilibrium Θ^* is E-stable if all eigenvalues of the Jacobian $DT(\Theta^*; \bar{S})$ have real parts strictly less than 1.

Proof. Suppose that $\mathbb{E}[Z_t' Z_t] = \mathbb{E}[(z_t z_t') \otimes I_3] \equiv M_Z < \infty$, $\mathbf{E}[Z_t' \eta_t^V] = 0$ and $\lim_{t\to\infty} R = \bar{R}$. Using stochastic recursive algorithm we can approximate (19) and (12) with the ordinary differential equation,

$$\frac{d\Theta}{d\tau} = g(\bar{R})^{-1} \Big(\lim_{t\to\infty} T(\hat{\Theta}_{t|t-1}; \bar{R}) - \hat{\Theta}_{t|t-1} \Big) M_Z$$

At equilibrium $\Theta = \Theta^*$, the Jacobian blocks associated with (Θ^V, Θ^X) are given by

$$DT^V(\Theta^*; \bar{R}) = (\Theta^{V*})' \otimes \Omega_2 \ + \ I_3 \otimes \big(\Omega_1 + \Omega_2 \Theta^{V*}\big), \tag{20}$$

$$DT^X(\Theta^*; \bar{R}) = H' \otimes \Omega_2 \ + \ I_3 \otimes \big(\Omega_1 + \Omega_2 \Theta^{V*}\big). \tag{21}$$

, where Θ^* is E-stable if all eigenvalues of (20)–(21) have real parts less than 1. Appendix A.5 shows that, under the regularity conditions stated above, the constant-gain learning recursions converge to a neighborhood of the MSV (with size governed by the gain g).

In standard decreasing-gain learning, belief revisions are driven by the innovation $V_t - Z_t \hat{\theta}_{t|t-1}$ scaled by the learning step size. In contrast, our constant-gain environment features an additional belief-uncertainty term in the one-step-ahead forecast, which enters the ALM directly. This covariance correction amplifies the endogenous response of V_t to shocks, thereby magnifying forecast errors and inducing larger belief updates (Figure 4). The resulting feedback loop—belief uncertainty $\rightarrow$ outcomes $\rightarrow$ forecast errors $\rightarrow$ belief updates—can prolong convergence and generate additional fluctuations relative to decreasing learning gain models. Adding to the determinacy result in Bullard and Mitra (2007), Figure 3 shows that stronger policy inertia can improve learnability by reducing the variation in the endogenous environment and weakening this feedback mechanism.

3 Extension

Background In this section, we consider an environment where agents strategically adjust their beliefs with private and public signals from the central bank announcements. In

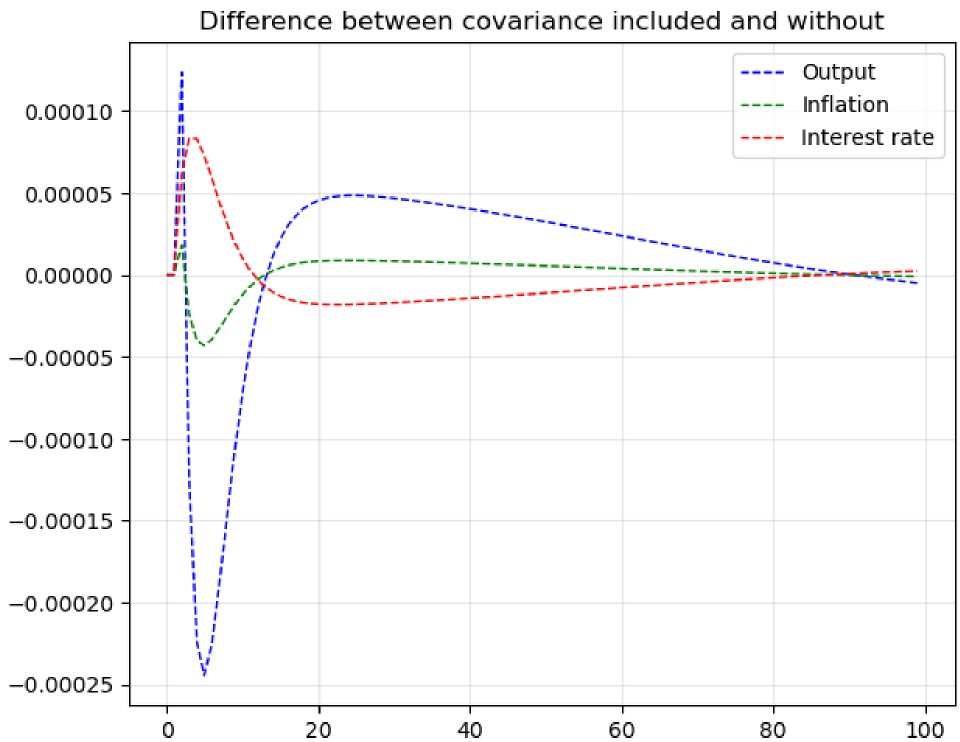
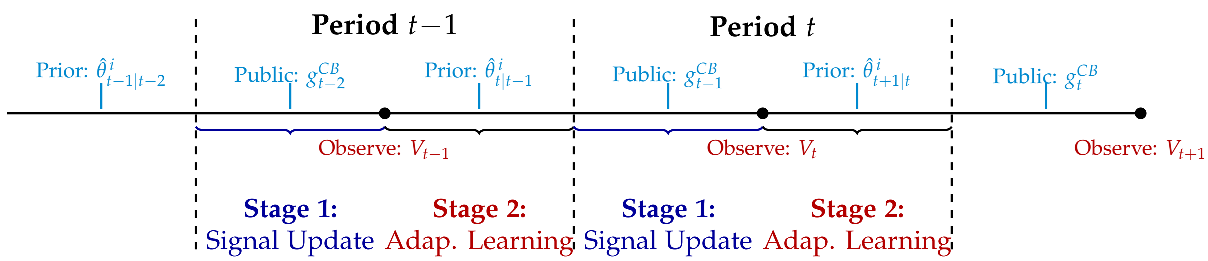


Figure 4: The figure plots the difference between the simulated IRFs with and without the covariance correction in the one-step-ahead forecast, following a 1% monetary policy (interest rate) shock.

essence, we combine the baseline model with Morris and Shin (2002); Hellwig and Veldkamp (2009); Gemmi and Valchev (2026) where agents first engage in eductive learning to forecast the average expectation and refine their expectation later upon state realization. The formulation of our environment can be seen as the special case of Evans et al. (2025) with adaptive dynamic of fixed level $k = \infty$ reasoning across all agents. Intuitively, the mixture of public and private signal can either catalyze or elongate the process of belief convergence depending on one's strategic motive and signals' relative precision.



Information Set and Flow of Belief Updates At the initial period ($t = 0$) agents are randomly endowed with heterogeneous prior beliefs $\hat{\theta}^{(i)}_{1|0} \sim N(\theta_1, \hat{P}_{1|0})$.

Each period consists of 2 stages. Agents enter with their prior beliefs $\hat{\theta}^{(i)}_{t|t-1}$ formed in the previous period's Stage 2. At the end of Stage 1, they receive a public signal from the central bank: $g^{(CB)}_{t-1} \sim N(\theta_t, \Sigma^{(CB)}_{t-1})$. We assume that the prior variance $\hat{P}_{t|t-1}$ and the public signal precision $\left(\Sigma^{(CB)}_{t-1}\right)^{-1}$ are common knowledge among all agents. Given these signals, agents participate into a forecasting game where they formulate their post-signal beliefs $\tilde{\theta}^{(i)}_{t|t-1}$ with the optimal forecasting rule, which incorporates both their private information and strategic considerations about others' beliefs.

All agents observe the same realization of V_t, which provides a common anchor for belief updating. Subsequently, agents update their post-signal beliefs $\tilde{\theta}^{(i)}_{t|t-1}$ with recursive learning algorithm to form posterior beliefs $\hat{\theta}^{(i)}_{t|t}$. These posteriors become the prior beliefs $\hat{\theta}^{(i)}_{t+1|t}$ in the next period's stage 1.

3.1 Stage 1 - Eductive Learning

Forecast Problem In Stage 1, agents engage in a forecast competition game. Given individual heterogeneous prior beliefs $\hat{\theta}^{(i)}_{t|t-1}$ and the public signal $g^{(CB)}_{t-1}$, agents formulate post-signal beliefs $\tilde{\theta}^{(i)}_{t|t-1}$ to forecast the current state θ_t.

Each individual's optimal forecast solves o the minimization problem:

$$\min_{\tilde{\theta}^{(i)}_{t|t-1}} \mathbf{E}\left[\left(\tilde{\theta}^{(i)}_{t|t-1} - \theta_t\right)^2 - \omega(\tilde{\theta}^{(i)}_{t|t-1} - \mu^{\theta}_{t|t-1})^2 \nmid \hat{\theta}^{(i)}_{t|t-1}, g^{CB}_{t-1}\right], \tag{22}$$

where $\mu^{\theta}_{t|t-1} \equiv \int \hat{\theta}^{(j)}_{t|t-1} dj$ denotes the average forecast , and $\omega \in (-1, 1)$ is the strategic parameter governing the relative importance of coordination versus differentiation. When $\omega \in (-1, 0)$ agents are coordinating with each other by choosing forecast closer to the average belief ($\mu^{\theta}_{t|t-1}$); whereas if $\omega \in (1, 0)$ they exhibit strategic substitution (differentiation) relying more heavily on their private information ($\hat{\theta}^{(i)}_{t|t-1}$) to distinguish themselves from others.

The first-order condition to (22) defines the optimal forecast :

$$\tilde{\theta}^{(i)}_{t|t-1} = \frac{1}{1-\omega}\mathbf{E}[\theta_t \mid \hat{\theta}^{(i)}_{t|t-1}, g^{CB}_{t-1}] - \frac{\omega}{1-\omega}\mathbf{E}[\mu^{\theta}_t \mid \hat{\theta}^{(i)}_{t|t-1}, g^{CB}_{t-1}]. \tag{23}$$

Equation (23) reveals that each agent's optimal post-signal belief is a weighted combination of (i) their expectation of the true state and (ii) their expectation of what others will forecast on average.

Eductive Learning To solve the optimal forecast problem in equation (23), agents must form expectations about both the true state and the average forecast of other agents. Following (Morris and Shin, 2002; Hellwig and Veldkamp, 2009; Evans and Honkapohja, 2001) and (Gemmi and Valchev, 2026), we adopt the eductive learning framework, in which agents engage in iterative higher-order reasoning to infer what others will forecast. This process is equivalent to k-level reasoning in the beauty contest game literature, where each "level" represents an additional round of strategic thinking about others' beliefs.

Suppose each agent adopts a linear forecast rule to update their prior belief upon receiving the public signal, using standard Bayesian (Kalman) weights:

$$E[\theta_t^{(i)}|g_{t-1}^{CB}, \hat{\theta}_{t|t-1}^{(i)}] = G_1^{(*)}\hat{\theta}_{t|t-1}^{(i)} + G_2^{(*)} g_{t-1}^{CB}, \tag{24}$$

where the weights depend on the relative precision of each signal: $G_1^{(*)} = \Big[(\hat{P}_{t|t-1})^{-1} + (\Sigma_{t-1}^{CB})^{-1}\Big]^{-1}(\hat{P}_{t|t-1})^{-1}$ and $G_2^{(*)} = \Big[(\hat{P}_{t|t-1})^{-1} + (\Sigma_{t-1}^{CB})^{-1}\Big]^{-1}(\Sigma_{t-1}^{CB})^{-1}$.
Because agent must forecast the average belief, where they conjecture that other agents $(j \neq i)$ follow a potentially different linear forecast rule with weights (G_1, G_2):

$$E[\theta_t^{(j)}|g_{t-1}^{CB}, \hat{\theta}_{t|t-1}^{(i)}] = G_1\hat{\theta}_{t|t-1}^{(j)} + G_2 g_{t-1}^{CB}. \tag{25}$$

Since individual agents are atomistic and cannot influence the cross-sectional average, and since prior beliefs are unbiased across agents $\int \hat{\theta}_{t|t-1}^{(j)} dj = \theta_t$, agent (i) can infer the expected average forecast as:

$$E[\mu_t^{\theta} \mid g_{t-1}^{CB}, \hat{\theta}_{t|t-1}^{(i)}] = G_1\mathbf{E}\big[\theta_t \mid g_{t-1}^{CB}, \hat{\theta}_{t|t-1}^{(i)}]\big] + G_2 g_{t-1}^{CB}. \tag{26}$$

Combine (25) and (26) with (23), agent (i)'s optimal forecast rule becomes:

$$E^{(NEW)}[\theta_t^{(j)}|g_{t-1}^{CB}, \hat{\theta}_{t|t-1}^{(i)}] = \underbrace{\frac{G_1^{(*)} - \omega G_1 G_1^{(*)}}{1-\omega}}_{G_1^{(NEW)}}\hat{\theta}_{t|t-1}^{(i)} + \underbrace{\frac{G_2^{(*)} - \omega G_1 G_2^{(*)} - \omega G_2}{1-\omega}}_{G_2^{(NEW)}} g_{t-1}^{CB}.$$

The new weights $G_1^{(NEW)}$ and $G_2^{(NEW)}$ represent the level-1 belief updating rule - agent (i)'s best response when they conjecture others use weights (G_1, G_2). However, if agent (i)

anticipates that others are also strategic, they must iterate further. At level k, agents conjecture others follow the level-$(k-1)$ rule and compute their own best response, yielding the recursive system:

$$
\begin{aligned}
G_1^{(k+1)} &= \frac{G_1^{(*)} - \omega G_1^{(k)} G_1^{(*)}}{1-\omega}, \\
G_2^{(k+1)} &= \frac{G_2^{(*)} - \omega G_1^{(k)} G_2^{(*)} - \omega G_2^{(k)}}{1-\omega}.
\end{aligned}
$$

Proposition 2 (Eductive Learning Equilibrium)**.** If agents engage in infinite iterations of this reasoning process ($k \to \infty$), the forecast weights converge to the eductive equilibrium values:

$$
\begin{aligned}
\lim_{k\to\infty} G_1^{(k)} &= \bar{G}_1 = \left[I(1-\omega) + \omega G_1^{(*)} \right]^{-1} G_1^{(*)}, \\
\lim_{k\to\infty} G_2^{(k)} &= \bar{G}_2 = \left(I - \omega \bar{G}_1 \right) G_2^{(*)},
\end{aligned}
$$

for any invertible matrix $G_1^{(*)}$.

3.2 Stage 2 – Adaptive Learning

After agents form post-signal beliefs $\tilde{\theta}^{(i)}_{t|t-1}$ in Stage 1, the endogenous state V_t is realized in Stage 2. Agents then update beliefs using standard Bayesian (Kalman) filtering. The resulting posteriors $(\hat{\theta}^{(i)}_{t|t}, \hat{P}^{(i)}_{t|t})$ become next period's priors $(\hat{\theta}^{(i)}_{t+1|t}, \hat{P}^{(i)}_{t+1|t})$ for Stage 1. Because all agents use the same eductive-equilibrium signal weights $(\bar{G}_1, \bar{G}_2)$ derived in Stage 1, Stage 2 updating is identical across agents. We therefore drop the individual index (i) for notational simplicity.

To simplify the analysis, we impose a symmetry assumption on private information. Agents are ex-ante identical and receive private signals of the same precision, but they do not internalize this commonality when forming higher-order beliefs in the forecasting game. That is, each agent treats others' forecasts as being based on distinct private information even though, in aggregate, the distribution of priors is symmetric. As a result, equilibrium updating is symmetric across agents, and we drop the individual superscript (i) throughout Stage 2 without loss of generality

In addition, we impose a *constant noise-to-signal ratio* structure following Sargent (1999); McCulloch (2025). This restriction isolates how relative signal precision and strategic incentives enter the learning dynamics, and it clarifies their respective roles in the E-stability conditions. Let the constant-gain (forgetting) parameter be $g = 1 - \lambda$ with $\lambda \in (0, 1)$. Assume parameter drift is proportional to the current posterior covariance,

$$R_{1,t} = \hat{P}_{t-1|t-1}\Big(\frac{1}{1-g} - 1\Big) = \hat{P}_{t-1|t-1}\Big(\frac{g}{1-g}\Big).$$

Then the prediction step implies

$$\hat{P}_{t|t-1} = \hat{P}_{t-1|t-1} + R_{1,t} = \frac{1}{1-g}\,\hat{P}_{t-1|t-1} = \frac{1}{\lambda}\,\hat{P}_{t-1|t-1}.$$

Next, assume the central bank provides a public signal about θ_t,

$$g^{CB}_{t-1} \sim \mathcal{N}(\theta_t, \hat{\Sigma}^{CB}_{t-1}), \qquad \hat{\Sigma}^{CB}_{t-1} = \gamma\, R_{1,t},$$

where $\gamma > 0$ scales the public-signal noise. Define the *relative noise ratio* (public vs. private) by

$$\kappa \;\equiv\; g\gamma \;=\; (1-\lambda)\gamma,$$

so that

$$(\hat{\Sigma}^{CB}_{t-1})^{-1} = \frac{1}{\kappa}\,(\hat{P}_{t|t-1})^{-1}.$$

Hence κ is the ratio of *public precision* to *private precision*: $\kappa < 1$ means the public signal is relatively more precise (more transparent policy), while $\kappa > 1$ means the public signal is relatively less precise (more opaque policy).

Under these assumptions, the Bayesian (rational) weights on private and public signals are

$$G_1^* = \frac{\kappa}{1+\kappa}, \qquad G_2^* = \frac{1}{1+\kappa}.$$

The eductive-equilibrium weights from Stage 1 are therefore

$$\bar{G}_1 = \frac{G_1^*}{1-\omega+\omega G_1^*} = \frac{\kappa}{1+\kappa-\omega}, \qquad \bar{G}_2 = (1-\omega\bar{G}_1)\,G_2^* = \frac{1-\omega}{1+\kappa-\omega}.$$

Substituting $(\bar{G}_1, \bar{G}_2)$ yields the post-signal belief and covariance,

$$\begin{aligned}
\tilde{\theta}_{t|t-1} &= \bar{G}_1\Big(\hat{\theta}_{t|t-1} + \rho\, g^{CB}_{t-1}\Big),\\
\tilde{P}_{t|t-1} &= \frac{1}{\lambda}\hat{P}_{t-1|t-1}\Big[(\bar{G}_1)^2 + \kappa(\bar{G}_2)^2\Big] = \frac{1}{\lambda}\hat{P}_{t-1|t-1}\frac{\kappa\big(\kappa+(1-\omega)^2\big)}{(1+\kappa-\omega)^2},
\end{aligned}$$

where

$$\rho \equiv \frac{\bar{G}_2}{\bar{G}_1} = \frac{1-\omega}{\kappa}.$$

Using the post-signal beliefs, agents forecast the future with

$$\mathbb{E}_{t-1}[V_t] = Z_t\,\tilde{\theta}_{t|t-1},$$
$$\mathbb{E}_{t-1}[V_{t+1}] = \mathbb{E}_{t-1}[Z_{t+1}]\,\tilde{\theta}_{t|t-1} \;+\; Z_t\,\tilde{P}_{t|t-1}\,\mathrm{vec}(J_V'),$$

where $Z_t = z_t' \otimes I_3$ is the expanded regressor matrix, and J_V is a selection matrix that extracts the relevant blocks so that $\tilde{P}_{t|t-1}\mathrm{vec}(J_V')$ corresponds to $\mathrm{Cov}(\tilde{\theta}^V_{t|t-1}, \tilde{\theta}_{t|t-1})$. And, beliefs evolve according to

$$\hat{\theta}_{t|t} = \tilde{\theta}_{t|t-1} + \hat{P}_{t|t} Z_t' (R_{2,t})^{-1} \Big(V_t - Z_t \tilde{\theta}_{t|t-1} \Big), \tag{27}$$

$$\hat{\Sigma}_{t|t} = \tilde{\Sigma}_{t|t-1} + Z_t' (R_{2,t})^{-1} Z_t, \tag{28}$$

where $f_t \equiv Z_t \tilde{P}_{t|t-1} Z_t' + R_{2,t}$ and $\hat{\Sigma}_{t|t} \equiv (\hat{P}_{t|t})^{-1}$.

Lemma 3.1 (Asymptotic convergence of the information matrix)**.** *Assume $R_{2,t} \equiv R_2 = \lambda/\alpha$ is constant and $\lim_{t\to\infty} \mathbb{E}[Z_t' Z_t] \equiv M_Z < \infty$. Then (28) converges to $\bar{\Sigma}$ given by*

$$\bar{\Sigma} = \frac{\alpha}{\lambda}\,\frac{1}{1 - a(\kappa,\omega)}\, M_Z,$$

and a necessary condition for $\{\hat{\Sigma}_{t|t}\}$ to converge to $\bar{\Sigma}$ is

$$a(\kappa,\omega) \;\equiv\; \frac{(1+\kappa-\omega)^2}{\kappa\big(\kappa + (1-\omega)^2\big)} \;<\; \frac{1}{\lambda}.$$

Let the public signal in matrix form be $G^{CB} = [G^{CB}_V, G^{CB}_X]$. Then the post-signal coefficients satisfy

$$\tilde{\Theta}^V = \bar{G}_1\big(\Theta^V + \rho G^{CB}_V\big), \qquad \tilde{\Theta}^X = \bar{G}_1\big(\Theta^X + \rho G^{CB}_X\big).$$

Rewrite the belief-updating process in stochastic recursive form

$$\hat{\theta}_{t|t} = \tilde{\theta}_{t|t-1} + \hat{P}_{t|t}(R_{2,t})^{-1} Z_t' \Big[Z_t\,\mathrm{vec}\big(T(\tilde{\Theta}_{t|t-1}) - \tilde{\Theta}_{t|t-1}\big) + Z_t\,\tilde{\Sigma}_{t|t-1}\mathrm{vec}(J_V') + Z_t \eta_t \Big],$$

where the T-map is defined as $T(\tilde{\Theta}_{t|t-1}) = \big[T^V(\tilde{\Theta}_{t|t-1}),\, T^X(\tilde{\Theta}_{t|t-1})\big]$, where

$$T^V(\tilde{\Theta}^V_{t|t-1}) = \Omega_1 \tilde{\Theta}^V_{t|t-1} + \Omega_2\big(\tilde{\Theta}^V_{t|t-1}\tilde{\Theta}^V_{t|t-1}\big) + \Omega_3,$$
$$T^X(\tilde{\Theta}^V_{t|t-1}, \tilde{\Theta}^X_{t|t-1}) = \Omega_1 \tilde{\Theta}^X_{t|t-1} + \Omega_2\big(\tilde{\Theta}^V_{t|t-1}\tilde{\Theta}^X_{t|t-1} + \tilde{\Theta}^X_{t|t-1} H\big) + \Omega_4 H.$$

Proposition 3 (E-stability)**.** Under the regularity conditions $M_Z < \infty$, $\mathbb{E}[Z_t \eta_t] = 0$, and

Lemma 3.1, local stability is governed by the ODE

$$\frac{d}{d\tau}\tilde{\Theta} \;=\; T(\tilde{\Theta}) - \tilde{\Theta}.$$

The Jacobian blocks satisfy

$$DT^V(\tilde{\Theta}) = (\tilde{\Theta}^V + \rho G_V^{CB})' \otimes (\Omega_2 \bar{G}_1^2) \;+\; I_3 \otimes \Big(\Omega_1 \bar{G}_1 + \Omega_2 \bar{G}_1^2 (\tilde{\Theta}^V + \rho G_V^{CB})\Big),$$
$$DT^X(\tilde{\Theta}) = I_3 \otimes \Big(\Omega_1 \bar{G}_1 + \Omega_2 \bar{G}_1^2 (\tilde{\Theta}^V + \rho G_V^{CB})\Big) \;+\; H' \otimes (\Omega_2 \bar{G}_1).$$

If a fixed point Θ^* exists and public signals are unbiased, $\mathbb{E}[G^{CB}] = \Theta^*$ (so $\mathbb{E}[G_V^{CB}] = \Theta^{V*}$ and $\mathbb{E}[G_X^{CB}] = \Theta^{X*}$), then Θ^* is locally E-stable if (i) all eigenvalues of $DT^V(\Theta^*)$ have real parts less than 1, and (ii) all eigenvalues of $DT^X(\Theta^*)$ have real parts less than 1.

Proof is in Appendix A.6.

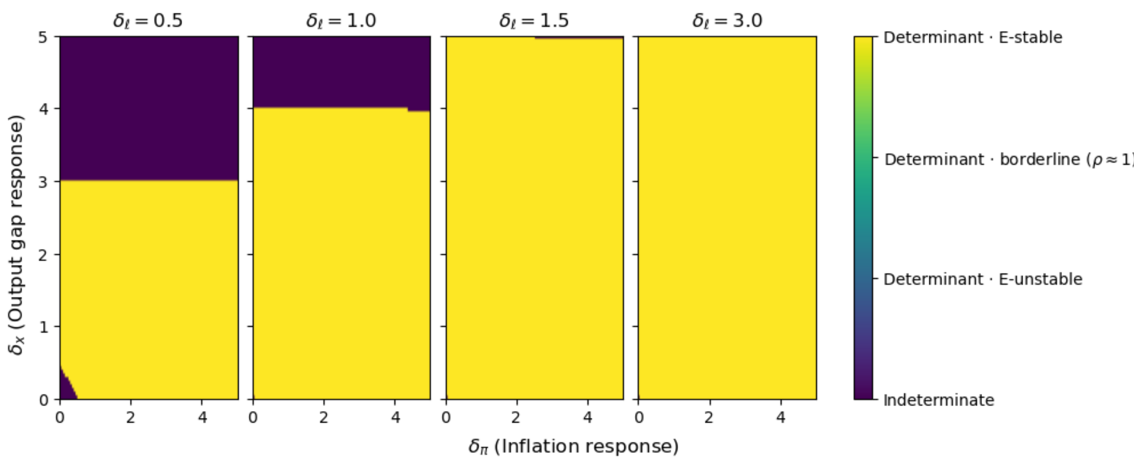


Figure 5: Determinacy and E-stability regions in the extension with strategic substitution and a precise public signal ($\omega = 0.5$, $\kappa = 0.5$). Each panel varies interest-rate smoothing δ_ℓ; colors indicate whether the REE is indeterminate, determinate but E-unstable, borderline E-stable, or strongly E-stable over the (δ_π, δ_x) policy-rule grid.

Comparative Statics Relative to Bullard and Mitra (2007), Figure 5 shows that strategic information acquisition can enlarge the set of policy rules that are learnable (E-stable) for a given degree of policy inertia. Intuitively, the equilibrium weights $(\bar{G}_1, \bar{G}_2)$ attenuate the sensitivity of post-signal beliefs to noisy public components, which makes belief trajectories less likely to drift into regions where the T-map is locally unstable. Consequently, E-stability can obtain even under more moderate policy inertia.

At the same time, transparency and strategic incentives matter for the magnitude of endogenous belief dispersion through Lemma 3.1. The stationary information matrix $\bar{\Sigma}$ scales with $a(\kappa, \omega)$, so (κ, ω) affect long-run belief tightness and therefore the strength of the belief-uncertainty feedback in equilibrium. In particular, lower public precision (higher κ) reduces $\bar{\Sigma}$ and increases belief dispersion, generating larger endogenous fluctuations. Moreover, $\partial a(\kappa, \omega)/\partial \kappa < 0$ for all $\omega \in (-1, 1)$, implying that increasing public precision (raising κ) strengthens long-run precision. The magnitude of this effect is larger under coordination motives (more negative ω), because equilibrium places relatively more weight on public information and changes in its precision have larger consequences for belief tightness.

3.3 Interaction between Strategic Information Acquisition and Policy Transparency

Calibration Before presenting the impulse responses, we briefly summarize the baseline calibration. The quarterly discount factor is set to $\beta = 0.99$ and the intertemporal elasticity parameter is normalized to $\sigma^{-1} = 1$. On the price-setting side, we set the slope parameter to $\xi = 0.005$ and the production curvature to $\theta = 0.3$. We allow for a small liquidity component in the Phillips curve by setting $\nu = 0.05$, and normalize the labor-supply elasticity parameter to $\chi = 1$. Notably, monetary policy follows the forward-looking rule in (3) with *moderate* level of inertia $\delta_L = 1$ and response coefficients $(\delta_\pi, \delta_x) = (0.7, 0.01)$. Finally, exogenous disturbances follow AR(1) processes with persistence parameters $\rho_u = 0.5$, $\rho_r = 0.5$, and $\rho_\epsilon = 0$.

Strategic motives and policy transparency. In the simulations, we set $\kappa = 0.5$ to represent a high-transparency (precise) communication regime and $\kappa = 2$ to represent a low-transparency (noisy) regime.

Figure 6 reports impulse responses to a 1% monetary policy (interest-rate) shock under strategic coordination ($\omega = -0.5$) and strategic substitution ($\omega = 0.5$). Two patterns stand out. First, lower transparency amplifies the volatility of impulse responses across all state variables. Second, this amplification is substantially stronger under strategic coordination than under strategic substitution. The intuition is that coordination motives shift equilibrium forecast weights toward the common (public) signal; when that signal is imprecise, agents move together in response to measurement error, raising aggregate volatility.

This mechanism is distinct from (and complementary to) the canonical Morris–Shin channel (Morris and Shin, 2002). In the static beauty-contest environment, equilibrium may overweight public information relative to the social optimum, so greater public precision is not necessarily welfare improving. In our constant-gain learning economy, policy

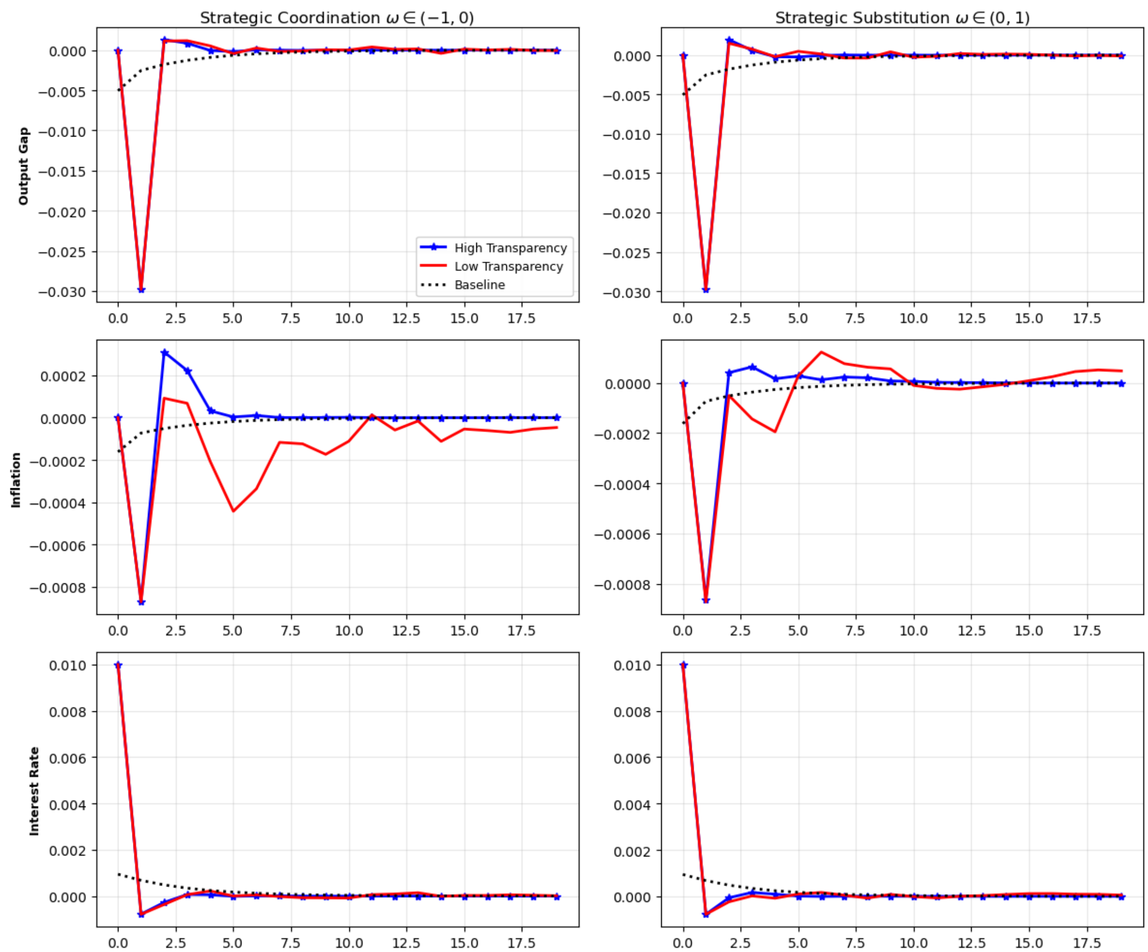


Figure 6: Impulse responses to a 1% monetary policy (interest-rate) shock under strategic coordination ($\omega = -0.5$; left column) and strategic substitution ($\omega = 0.5$; right column). Blue lines show the high-transparency regime ($\kappa = 0.5$); red lines show the low-transparency regime ($\kappa = 2$) with the covariance correction active. The black dashed line reports the rational-expectations benchmark (no strategic interaction and no covariance correction).

precision plays an additional dynamic role: improving the informativeness of the common signal reduces forecast errors and belief dispersion, weakens the covariance correction that enters one-step-ahead forecasts (and thus the ALM), and limits shock amplification through the belief-updating feedback. Consequently, noisier policy communication generates larger forecast errors and induces larger belief revisions, prolonging the adjustment back to steady state.

Finally, Table 1 shows the contribution of belief uncertainty differs across strategic regimes. Under strategic substitution, agents place relatively more weight on private

	Coordination ($\omega = -0.5$)				Substitution ($\omega = 0.5$)			
	High transp ($\kappa = 0.5$)		Low transp ($\kappa = 2$)		High transp ($\kappa = 0.5$)		Low transp ($\kappa = 2$)	
	With Cov	No Cov	With Cov	No Cov	With Cov	No Cov	With Cov	No Cov
Output gap	0.0029	0.0030	0.0036	0.0032	0.0030	0.0030	0.0031	0.0031
Inflation	1.1e-4	1.0e-4	2.1e-4	1.6e-4	1.1e-4	9.8e-5	1.3e-4	1.4e-4
Interest rate	7.8e-4	7.8e-4	9.2e-4	9.0e-4	7.9e-4	8.0e-4	8.1e-4	8.3e-4

Table 1: Each entry reports the RMS deviation of the simulated impulse response function (IRF) from the baseline over the IRF horizon. Columns split results by strategic motive (coordination ($\omega = -0.5$) vs. substitution ($\omega = 0.5$)), policy transparency (high ($\kappa = 0.5$) vs. low ($\kappa = 2$)), and whether the covariance (belief-uncertainty) correction is included.

information, so ignoring the covariance channel can understate endogenous fluctuations even when communication is relatively precise. Under strategic coordination, beliefs are more tightly anchored to the common signal, which compresses dispersion and attenuates the covariance-driven component of fluctuations—even though aggregate volatility can still rise sharply when the common signal itself becomes noisy.

4 Conclusion

This paper studies how monetary-policy communication affects macroeconomic stability when expectations are formed through (i) perpetual learning with time-varying perceived laws of motion and (ii) strategic information acquisition. We embed constant-gain learning and Kalman-filter updating in a New Keynesian environment in which agents combine a private estimate of the economy with a public policy signal. The public signal is used strategically: equilibrium weights reflect agents' incentives to coordinate or differentiate their forecasts. A key feature of the learning environment is that belief uncertainty matters for outcomes, because a covariance correction enters into the economy through one-step-ahead forecasts. This channel creates an endogenous feedback between belief uncertainty, realized macroeconomic outcomes, and subsequent belief updating.

Our main theoretical contribution is to connect two literatures that are often studied separately: the learning literature and the strategic information-acquisition literature. We develop a tractable framework to analyze how the precision of policy announcements interacts with strategic motives and belief uncertainty in shaping determinacy and E-stability. Relative to benchmark learning models in which belief revisions are driven only by forecast innovations scaled by a learning gain, our framework features an additional covariance-driven component. This term can amplify the endogenous response to shocks by affecting one-step-ahead forecasts, thereby increasing volatility and potentially slowing convergence.

Quantitatively, the model delivers two main implications. First, strategic information acquisition can materially enlarge the set of monetary-policy rules that are learnable (E-stable) even when policy inertia is only moderate ($\delta_L = 1$). In this sense, macroeconomic stability is not determined by the policy rule alone: it also depends on the information environment in which expectations are formed. When agents internalize the forecasting externality, equilibrium signal weights attenuate the effective responsiveness of beliefs to transitory disturbances, making belief trajectories less likely to drift into locally unstable regions. As a result, policies that would be difficult to learn in standard constant-gain setups can become learnable once strategic interaction is taken into account.

Second, policy transparency remains central for short-run stabilization. A more precise public signal anchors higher-order beliefs and reduces belief dispersion, weakening the covariance correction that enters one-step-ahead forecasts and, through the ALM, dampening endogenous amplification of monetary shocks. This stabilizing role of transparency is strongest under strategic coordination, where equilibrium forecasts place greater weight on the common signal: improving its precision tightens beliefs around the common anchor and limits the scope for joint overreaction to noise. Overall, the results suggest that communication design complements conventional policy-rule design, particularly when agents' information choices are strategic.

Finally, our results refine the transparency tradeoff emphasized in the Morris–Shin framework, where greater public-signal precision can be socially harmful in a static coordination environment. In our setting, transparency operates through an additional dynamic channel: by shaping belief dispersion and the covariance-feedback term, policy precision affects not only contemporaneous coordination on signals but also the propagation of shocks through learning. Within our calibrated environment, we find a clear stabilizing force: when coordination motives are strong, sufficiently precise announcements mitigate belief over-adjustment by improving the informativeness of the common signal and dampening the covariance-feedback mechanism. A practical implication is that announcement precision can be an effective stabilization tool—especially in environments where incentives favor coordination—because clearer communication strengthens the common anchor and reduces learning-driven amplification.

A Appendix A: Proofs

A.1 Determinacy Condition

In this subsection, we follow the approach in Woodford (2003) to find the necessary and sufficient conditions for determinacy.

- Denote the vector of endogenous state variables as $V_t = [x_t, \pi_t, i_t]'$, re-write 1, 2, **??** and 3 as the following matrix form:

$$\begin{aligned} V_t &= \begin{bmatrix} \sigma^{-1}(\beta^{-1}-1) \\ 0 \\ (\beta^{-1}-1) \end{bmatrix} + \begin{bmatrix} 0 & 0 & \sigma^{-1} \\ \xi\kappa & 0 & \varsigma \\ 0 & \frac{\lambda_x}{\xi(\kappa-\sigma\varsigma)} & \phi\rho_u \end{bmatrix} \mathbf{E}^*_{t-1}V_t \\ &+ \begin{bmatrix} 1 & \sigma^{-1} & 0 \\ 0 & \beta & 0 \\ 0 & 0 & 0 \end{bmatrix} \mathbf{E}^*_{t-1}V_{t+1} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \delta_L \end{bmatrix} V_{t-1} + \begin{bmatrix} \sigma^{-1} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r^n_t \\ u_t \\ \epsilon_t \end{bmatrix}. \\ &= \Omega_0 + \Omega_1 \mathbf{E}_{t-1}V_t + \Omega_2 \mathbf{E}_{t-1}V_{t+1} + \Omega_3 V_{t-1} + \varepsilon^V_t. \end{aligned} \tag{29}$$

- And the vector of exogenous state variables is $X_t = [r_t, u_t, \epsilon_t]'$ following an AR(1) stochastic process so that

$$\begin{aligned} X_t &= \begin{bmatrix} \rho_u & 0 & 0 \\ 0 & \rho_r & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} r_{t-1} \\ u_{t-1} \\ \epsilon_{t-1} \end{bmatrix} + \begin{bmatrix} \varepsilon^{X_r}_t \\ \varepsilon^{X_u}_t \\ \varepsilon^{X_\epsilon}_t \end{bmatrix} \\ &= HX_{t-1} + \varepsilon^X_t \end{aligned} \tag{30}$$

  , note that ε^X contain mean-zero i.i.d shocks that are also independent of ε^V.

Suppose that innovation of one-step ahead forecast error is : $\mathbf{E}_t V_{t+1} - \mathbf{E}_{t-1}V_{t+1} = \eta_t$. Then the above system of equations can be expressed in the following form:

$$A\mathbf{E}_{t-1}V_t = C + DV_{t-1} + EX_{t-1} + \eta_t$$

where,

$$A = \begin{bmatrix} \sigma^{-1}\delta_x - 1 & \sigma^{-1}(\delta_\pi - 1) & 0 \\ -\xi\left[\kappa + (\xi - \kappa\sigma^{-1})\delta_x\right] & -\left(\beta + \xi\left[\kappa\xi^{-1} + (\xi - \kappa\sigma^{-1})\delta_\pi\right]\right) & 0 \\ -\delta_x & -\delta_\pi & 1 \end{bmatrix},$$

$$D = \begin{bmatrix} -1 & 0 & -\sigma^{-1}\delta_i \\ 0 & -1 & \xi(\xi - \kappa\sigma^{-1})\delta_i \\ 0 & 0 & \delta_i \end{bmatrix}.$$

$$C = \begin{bmatrix} \sigma^{-1}(\beta^{-1} - 1) \\ \xi\kappa\left(\sigma^{-1}(\beta^{-1} - 1)\right) \\ 0 \end{bmatrix}.$$

Lemma A.1. *Denote $B = (A^{-1})C$. Let $B_0 = det(B)$, $B_1 = \frac{1}{2}[(Tr(B))^2 - Tr(B^2)]$ and $B_2 = -Tr(B)$. Woodford (2003) shows that the determinacy conditions are:*

$$\begin{aligned} 1 + B_0 + B_1 + B_2 &> 0 \\ -1 + B_0 - B_1 + B_2 &< 0 \end{aligned}$$

, and either

$$(B_0)^2 - B_0 B_2 + B_1 - I > 0$$

or

$$\begin{aligned} (B_0)^2 - B_0 B_2 + B_1 - I &< 0 \\ det(B_2) &> 3. \end{aligned}$$

The first 2 conditions imply that Talyor rule is the necessary condition for determinacy, but the remaining conditions require sufficiently strong policy inertia, that is

$$\delta_L \;\geq\; \delta_x \left\{ \frac{\beta + \zeta\xi - 1}{\kappa\,\zeta} \;-\; \frac{\beta + \zeta\xi}{\zeta\,\xi\,\sigma} \right\} \;+\; 1 \;+\; \frac{\beta}{\zeta\,\xi}.$$

in the first case, or

$$\delta_L \;>\; \max\left\{ \frac{\delta_x(\beta + \zeta\xi - 1)}{\kappa\,\zeta} + (1 - \delta_\pi),\; \frac{\delta_x(1 + \beta + \zeta\xi) + \delta_\pi(\kappa\zeta - \zeta\xi\sigma) - \kappa\zeta - 2\sigma - 2\beta\sigma}{\kappa\zeta + 2\sigma + 2\beta\sigma} \right\}.$$

in the last case.

A.2 Minimal State Variable Solution (MSV)

Proof shown here is a direct adaptation of Evans and Honkapohja (2001) Chapter 10.

- Let
$$\eta_t^{\mathfrak{x}} = \mathbf{E}_t \mathfrak{x}_{t+1} - \mathbf{E}_{t-1}\mathfrak{x}_{t+1} \qquad \text{for} \qquad \mathfrak{X} = x, \pi, i$$

- Since
$$\begin{bmatrix} x_t \\ \pi_t \\ i_t \end{bmatrix} - \mathbf{E}_{t-1} \begin{bmatrix} x_t \\ \pi_t \\ i_t \end{bmatrix} = \begin{bmatrix} \sigma^{-1} r_t^n \\ u_t \\ \epsilon_t \end{bmatrix}$$

  , it follows that

$$\begin{bmatrix} x_{t+1} \\ \pi_{t+1} \\ i_{t+1} \end{bmatrix} - \mathbf{E}_{t-1} \begin{bmatrix} x_{t+1} \\ \pi_{t+1} \\ i_{t+1} \end{bmatrix} = \begin{bmatrix} \sigma^{-1} r_{t+1}^n \\ u_{t+1} \\ \epsilon_{t+1} \end{bmatrix} + \begin{bmatrix} \eta_t^x \\ \eta_t^\pi \\ \eta_t^i \end{bmatrix}$$

- Write equation 1, 2 and **??** in matrix form

$$\mathbf{Q} \begin{bmatrix} x_t \\ \pi_t \\ i_t \\ r_t^n \\ u_t \\ \epsilon_t \\ x_t^l \end{bmatrix} = \mathbf{C} + \mathbf{B} \begin{bmatrix} x_{t-1} \\ \pi_{t-1} \\ i_{t-1} \\ r_{t-1}^n \\ u_{t-1} \\ \epsilon_{t-1} \\ x_{t-1}^l \end{bmatrix} + \mathbf{K}\eta_t \tag{31}$$

  , where
$$\mathbf{Q} = \begin{bmatrix} 1 & -\sigma^{-1} & \sigma^{-1} & \sigma^{-1} & 0 & 0 & 0 \\ -\varsigma\kappa & 1 & -\varsigma\varphi & -\varsigma\kappa\sigma^{-1} & -1 & \varsigma\varphi & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & \delta_L \\ 0 & 0 & 0 & \rho_r & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \rho_u & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \rho_\epsilon & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

-
$$\mathbf{C} = \begin{bmatrix} \sigma^{-1}(\beta^{-1} - 1) & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}'$$

- $$\mathbf{B} = \begin{bmatrix} 1 & 0 & \sigma^{-1} & \sigma^{-1} & 0 & \sigma^{-1} & 0 \\ 0 & \beta & 0 & 0 & -\beta & 0 & 0 \\ \delta_x & \delta_\pi & 0 & -\delta_x\sigma^{-1} & \delta_\pi & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

- Let $x_t^l = x_{t-1}$. Denote free variables as $y_t^1 = [x_t, \pi_t, i_t]'$ and predetermined variables as $y_t^2 = [r_t, u_t, \epsilon_t, x_t^l]'$

- Re-write the above matrix as :

$$\begin{bmatrix} Q^{11}, Q^{12} \\ Q^{21}, Q^{22} \end{bmatrix} \begin{bmatrix} y_t^1 \\ y_t^2 \end{bmatrix} = \mathbf{C} + \begin{bmatrix} B^{11}, B^{12} \\ 0, \mathbf{I} \end{bmatrix} \begin{bmatrix} y_{t+1}^1 \\ y_{t+1}^2 \end{bmatrix} + \mathbf{K}\eta_t$$

- Suppose $\mathbf{Q}$ is invertible, such that

$$\mathbf{Q}^{-1}\mathbf{B} = \mathbf{J}$$
$$\Rightarrow \mathbf{QJ} = \Lambda\mathbf{Q}$$

, where Λ is a diagonal matrix with entry Λ_1 and Λ_2.

- It follows that:

$$Q^{11}y_t^1 + Q^{12}y_t^2 = \mathbf{C}_1 + \Lambda_1\left(Q^{11}y_{t+1}^1 + Q^{12}y_{t+1}^2\right) + \mathbf{K}_1\eta_t$$

- Let $p_t = Q^{11}y_t^1 + Q^{12}y_t^2$, the above can be written as

$$p_t = \mathbf{C} + \Lambda_1 p_{t+1} + \mathbf{K}\eta_t$$

, where this is a stationary process for any $|\Lambda_1| < 1$ if

$$p_t = \mathbf{C} + \mathbf{K}\eta_t$$

- As such

$$y_t^1 = (Q^{11})^{-1}\mathbf{C} - (Q^{11})^{-1}Q^{12}y_t^2 + (Q^{11})^{-1}\mathbf{K}\eta_t$$

- Since

$$y_t^2 = \mathbf{G}_1 y_{t-1}^1 + \mathbf{G}_2 y_{t-1}^2 + \nu_t$$

, where

$$\mathbf{G}_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\mathbf{G}_2 = \begin{bmatrix} \rho_r & 0 & 0 & 0 \\ 0 & \rho_u & 0 & 0 \\ 0 & 0 & \rho_\epsilon & 0 \end{bmatrix}$$

. Note that $\rho_\epsilon = 0$ in the model.

- Combine the above to show that

$$y_t^1 = (Q^{11})^{-1}\mathbf{C} - (Q^{11})^{-1}Q^{12}\left(\mathbf{G}_1 y_{t-1}^1 + \mathbf{G}_2 y_{t-1}^2 + \nu_t\right) + \nu_t$$

Modify the above with 3 is straightforward with the following changes

- Let $\tilde{y}_t^2 = [r_t, u_t, \epsilon_t, i_t^l]'$
- $\mathbf{Q}^{21} = \begin{bmatrix} 1 & -\sigma^{-1} & \sigma^{-1} \\ -\varsigma\kappa & 1 & \varsigma\kappa \\ 0 & 0 & 1 \end{bmatrix}$
- $\mathbf{Q}^{21} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$
- $\mathbf{B}^{11} = \begin{bmatrix} 1 & 0 & \sigma^{-1} \\ 0 & \beta & 0 \\ \varphi_x & \varphi_\pi & 0 \end{bmatrix}$
- $\mathbf{B}^{12} = \begin{bmatrix} -\sigma^{-1} & 0 & \sigma^{-1} & 0 \\ 0 & -\beta & 0 & 0 \\ -\varphi_x\sigma^{-1} & -\varphi_\pi & 0 & 0 \end{bmatrix}$

A.3 Rational Expectation Equilibrium

This subsection shows that the rational expectation equilibrium of the economic model should have the following minimal state variable solution (MSV).

Let

$$V_t = \begin{bmatrix} x_t \\ \pi_t \\ i_t \end{bmatrix}, \qquad X_t = \begin{bmatrix} r_t^n \\ u_t \\ \varepsilon_t \end{bmatrix}, \qquad X_t = HX_{t-1} + e_t^X,$$

where $H = \text{diag}(\rho_r, \rho_u, \rho_\varepsilon)$ and $e_t^X = (\eta_{r,t}, \eta_{u,t}, \eta_{\varepsilon,t})'$ collects the exogenous shocks.

The structural equations are

$$x_t = \sigma^{-1}(\beta^{-1} - 1) - \sigma^{-1}E_{t-1}i_t + \sigma^{-1}E_{t-1}\pi_{t+1} + E_{t-1}x_{t+1} + \sigma^{-1}r_t^n, \tag{32}$$

$$\pi_t = \beta E_{t-1}\pi_{t+1} + \xi\left[\kappa x_t + \zeta i_t\right] + u_t, \tag{33}$$

$$i_t = \zeta_\pi E_{t-1}\pi_{t+1} + \zeta_x E_{t-1}x_{t+1} + \zeta_i i_{t-1} + \varepsilon_t. \tag{34}$$

These can be written compactly as

$$V_t = \Omega_0 + \Omega_1 E_{t-1}V_t + \Omega_2 E_{t-1}V_{t+1} + \Omega_3 V_{t-1} + \Omega_4 X_t, \tag{35}$$

with

$$\Omega_0 = \begin{bmatrix} \sigma^{-1}(\beta^{-1} - 1) \\ 0 \\ 0 \end{bmatrix}, \quad \Omega_1 = \begin{bmatrix} 0 & 0 & -\sigma^{-1} \\ \xi\kappa & 0 & \xi\zeta \\ 0 & 0 & 0 \end{bmatrix},$$

$$\Omega_2 = \begin{bmatrix} 1 & \sigma^{-1} & 0 \\ 0 & \beta & 0 \\ \zeta_x & \zeta_\pi & 0 \end{bmatrix}, \quad \Omega_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \zeta_i \end{bmatrix}, \quad \Omega_4 = \begin{bmatrix} \sigma^{-1} & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

We treat V_t as the vector of *free* variables and X_t as *predetermined*. Define the enlarged predetermined vector

$$Y_t = \begin{bmatrix} X_t \\ V_{t-1} \\ X_{t-1} \end{bmatrix},$$

so that (35) becomes

$$V_t = \Omega_0 + \Omega_1 E_{t-1}V_t + \Omega_2 E_{t-1}V_{t+1} + \left[0,\ \Omega_3,\ \Omega_4 H\right] Y_t + \Omega_4 e_t^X. \tag{36}$$

The law of motion for Y_t can be written as

$$Y_{t+1} = RV_t + SY_t + u_{2,t+1}, \qquad u_{2,t+1} = \begin{bmatrix} e^X_{t+1} \\ 0 \\ 0 \end{bmatrix}, \tag{37}$$

where

$$R = \begin{bmatrix} 0 \\ I_3 \\ 0 \end{bmatrix}, \qquad S = \begin{bmatrix} H & 0 & 0 \\ 0 & 0 & 0 \\ I_3 & 0 & 0 \end{bmatrix}.$$

Define the innovation (news term) in expected future V_{t+1} as

$$\eta^V_t := E_t V_{t+1} - E_{t-1} V_{t+1}. \tag{38}$$

From (35) we have

$$V_t - E_{t-1} V_t = \Omega_4 e^X_t,$$

and hence

$$V_{t+1} - E_{t-1} V_{t+1} = \Omega_4 e^X_{t+1} + \eta^V_t.$$

Equivalently,

$$E_{t-1} V_{t+1} = V_{t+1} - \Omega_4 e^X_{t+1} - \eta^V_t. \tag{39}$$

Substituting (39) into (36) yields

$$\begin{aligned} V_t &= \Omega_0 + \Omega_1 \big(V_t - \Omega_4 e^X_t\big) + \Omega_2 \big(V_{t+1} - \Omega_4 e^X_{t+1} - \eta^V_t\big) + \big[0,\ \Omega_3,\ \Omega_4 H\big] Y_t + \Omega_4 e^X_t \\ &= \Omega_0 + \Omega_1 V_t + \Omega_2 V_{t+1} + \big[0,\ \Omega_3,\ \Omega_4 H\big] Y_t \\ &\quad + (\Omega_4 - \Omega_1 \Omega_4) e^X_t - \Omega_2 \Omega_4 e^X_{t+1} - \Omega_2 \eta^V_t. \end{aligned} \tag{40}$$

Stacking (40) and (37) we obtain

$$\underbrace{\begin{bmatrix} I_3 - \Omega_1 & \begin{bmatrix} 0 & -\Omega_3 & -\Omega_4 H \end{bmatrix} \\ R & S \end{bmatrix}}_{=:M} \begin{bmatrix} V_t \\ Y_t \end{bmatrix} = \begin{bmatrix} \Omega_0 \\ 0 \end{bmatrix} + \underbrace{\begin{bmatrix} \Omega_2 & 0 \\ 0 & I_9 \end{bmatrix}}_{=:D} \begin{bmatrix} V_{t+1} \\ Y_{t+1} \end{bmatrix} + \Xi_t, \tag{41}$$

where

$$\Xi_t = \begin{bmatrix} (\Omega_4 - \Omega_1 \Omega_4) e^X_t - \Omega_2 \Omega_4 e^X_{t+1} - \Omega_2 \eta^V_t \\ u_{2,t+1} \end{bmatrix}.$$

Let

$$Z_t := \begin{bmatrix} V_t \\ Y_t \end{bmatrix}.$$

Lemma A.2. *Suppose M in* (41) *is invertible and that the matrix*

$$J := M^{-1}D$$

is diagonalizable with all eigenvalues inside the unit circle, $|\lambda_i(J)| < 1$. Then there exists a (unique) stationary rational expectations equilibrium of the form

$$V_t = \Theta^0 + \Theta^1 V_{t-1} + \Theta^2 X_{t-1} + \xi_t^V, \tag{42}$$

where $\Theta^0 \in \mathbb{R}^3$, $\Theta^1, \Theta^2 \in \mathbb{R}^{3\times 3}$, and ξ_t^V is a zero-mean martingale difference sequence, affine in the current and future shocks $(e_t^X, e_{t+1}^X, \eta_t^V)$.

Proof. From (41) we have

$$Z_t = M^{-1}\begin{bmatrix} \Omega_0 \\ 0 \end{bmatrix} + JZ_{t+1} + M^{-1}\Xi_t.$$

Iterating forward s steps gives

$$Z_t = \sum_{j=0}^{s-1} J^j M^{-1}\begin{bmatrix} \Omega_0 \\ 0 \end{bmatrix} + J^s E_t Z_{t+s} + \sum_{j=0}^{s-1} J^j M^{-1} E_t \Xi_{t+j},$$

using the law of iterated expectations and the martingale difference properties of (e_t^X, η_t^V).

By assumption, J is diagonalizable, $J = Q\Lambda Q^{-1}$, with $|\lambda_i(\Lambda)| < 1$. Hence $J^s \to 0$ as $s \to \infty$, so in the limit

$$Z_t = K_0 + \sum_{j=0}^{\infty} J^j K_1 \varepsilon_{t+1-j},$$

for suitable constant matrices K_0, K_1 and a stacked shock vector ε_{t+1-j} collecting $e_t^X, e_{t+1}^X, \eta_t^V$. The first three components of Z_t correspond to V_t, so they admit an affine representation in (X_{t-1}, V_{t-1}) plus a zero-mean innovation term. In particular,

$$V_t = \Theta^0 + \Theta^1 V_{t-1} + \Theta^2 X_{t-1} + \xi_t^V$$

for some matrices $\Theta^0, \Theta^1, \Theta^2$ and an innovation ξ_t^V that is linear in the current and future shocks and has $E_{t-1}\xi_t^V = 0$. This yields (42) and completes the proof. □

A.4 Correspondence of Kalmen Filter and Recursive Least Squared Learning Dynamics

Ljung et al. (1992); Sargent (1999) shows that learning with Kalman Filter closely tie to recursive least squares learning when the co-variances are proportionally related to each other. We begin by defining the recursive least squared learning problem.

At t, agent estimate the RLS estimator θ weighted with forgetting parameter $\lambda \in (0,1]$ and parameter α that minimizes the loss function:

$$L_t(\theta) = \sum_{s=1}^{t} \lambda^{t-s} \alpha_s \big(V_s - (z_s \otimes I_3)\theta\big)^2,$$

where the optimal solution yields the following recursive equations:

$$\hat{\theta}_t = \hat{\theta}_{t-1} + g_t \alpha_t R_t^{-1} Z_t' \big[V_t - Z_t \hat{\theta}_{t-1}\big] \tag{43}$$

$$R_t = R_{t-1} + g_t \alpha_t \big(Z_t' Z_t^{-} R_{t-1}\big) \tag{44}$$

$$g_t = \Big(\sum_{s=1}^{t} \lambda^{t-s}\Big)^{-1} = \begin{cases} \dfrac{1-\lambda}{1-\lambda^t}, & |\lambda| < 1, \\ \dfrac{1}{t}, & |\lambda| = 1. \end{cases} \tag{45}$$

For any $\lambda \in (0,1)$ and $\alpha \in \mathbb{R}^+$, the Kalman Filter system with equation (8), (9) and (10) is equivalent to recursive least squared (RLS) learning if

$$R_{1,t} = \frac{1-\lambda}{\lambda} \hat{\Sigma}^{\theta}{}_{t|t-1}, \qquad R_{2,t} = \frac{\lambda}{\alpha} \tag{46}$$

A.5 Local Convergence Result

In this appendix, we sketch the proof following the steps in (Evans and Honkapohja, 2001) to show the asymptotic convergence of the baseline model.

Let $S_{t-1} = R_t$ and re-write the recursive equations (11) and (12) as

$$\begin{aligned} H_\theta(\phi_{t-1}, X_t) = g(S_{t-1})^{-1} Z_t' \Big\{ & Z_t\big(T(\hat{\theta}_{t|t-1}, S_{t-2}) - \theta_{t-1}\big) \\ & + Z_t\big((1-g)S_{t-2}\big)^{-1} + Z_t \eta_t^V \Big\}, \end{aligned}$$

$$H_S(\phi_{t-1}, X_t) = S_{t-1} + (1-g)\Big(Z_{t+1}' Z_{t+1} - S_{t-1}\Big).$$

, where

$$\phi_t = \begin{bmatrix} \hat{\theta}_{t|t-1} \\ S_t \end{bmatrix}. \qquad X_t = \begin{bmatrix} \mathbb{E}_{t-1}[Z_{t+1}] \\ \mathbb{E}_{t-1}[Z_t] \\ \eta_t^V \end{bmatrix}.$$

Then we can express this system in standard form

$$\phi_t = \phi_{t-1} + g\, H(\phi_{t-1}, X_t),$$

, where $H(\phi_{t-1}, X_t) = \begin{bmatrix} H_\theta(\phi_{t-1}, X_t) \\ H_S(\phi_{t-1}, X_t) \end{bmatrix}$.

Suppose the following regularity conditions hold, that is (i) $\theta \in \mathbb{R}$ and (ii)$S \in (\xi, \infty)$, where $\xi > 0$, (iii) $Z_t \in [\underline{Z}, \bar{Z}]$ for all t so that $\mathbb{E}\, Z_t' Z_t = M_Z < \infty$, (iv) η_t^V has compact support, and $\mathbb{E}\left[Z_t \eta_t^{V\prime}\right] = 0$.

Define the ODE $\frac{d\phi}{d\tau} = h(\phi)$, and rewrite $H(\theta, S)$ as $h(\phi) = \begin{bmatrix} h_\theta(\theta, S) \\ h_S(\theta, S) \end{bmatrix}$, where

$$\begin{aligned} h_\theta(\theta, S) = S^{-1}\Big\{ & \Omega_0 + (\Omega_1 + \Omega_2 + \Omega_2\Theta^V - I_3)\Theta^c \\ & + \left[\Omega_3 + (\Omega_1 + \Omega_2\Theta^V - I_3)\Theta^V\right] M_V \\ & + \left[\Omega_1 + \Omega_4 H + (\Omega_2(\Theta^V + H) - I_3)\Theta^x\right] M_K + \Omega_2(1-\lambda)S^V \Big\}, \\ h_S(\theta, S) = {} & g M_Z - S. \end{aligned}$$

Proposition 4. For ϕ^* to be any locally stable point, θ_t can be approximated by

$$\phi_t \sim \mathcal{N}\left(\phi^*, (1-\lambda)C\right), \qquad R^* = \sum_{k=0}^{\infty} \mathrm{cov}\Big(H(\phi, X_k), H(\phi, X_0)\Big) \qquad C = \int_0^\infty e^{sB} R^* e^{sB'}\, ds.$$

, where $B = D_\phi h(\phi)$.

The blocks of the Jacobian

$$B = D_\phi h(\phi^*) = \begin{bmatrix} D_\theta h_\theta & D_S h_\theta \\ D_\theta h_S & D_S h_S \end{bmatrix}.$$

$$D_S h_\theta = \begin{bmatrix} 0 & 0 & 0 \\ 0 & I_3 \otimes \Omega_2 g & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

, where the only entry is non-zero because $\bar{S} = gM_z > 0$ under constant gain learning.

, and

$$D_\theta h_\theta = \begin{bmatrix} I_3 \otimes \left(\Omega_1 + \Omega_2(I_3 + \Theta^V)\right) & \Theta^c \otimes \Omega_2 & 0 \\ 0 & \Theta^V \otimes \Omega_2 + I_3 \otimes \left(\Omega_1 + \Omega_2 \Theta^V\right) & 0 \\ 0 & \Theta^x \otimes \Omega_2 & I_3 \otimes \left(\Omega_1 + \Omega_2(\Theta^V + H)\right) \end{bmatrix}.$$

, and

$$D_\theta h_S = 0_{21\times 21}, \qquad D_S h_S = -I_{21\times 21}.$$

Next, we analyze the asymptotic convergence of the covariance matrix.

$$R^* = \begin{bmatrix} R_{\theta\theta} & 0 \\ 0 & R_{ss} \end{bmatrix}.$$

, where

$$R_{\theta\theta} = \sum_{k=-\infty}^{\infty} \operatorname{Cov}\big(H_\theta(\phi, X_k), H_\theta(\phi, X_0)\big) = S^{-1} M_{\mathcal{Z}}\, \Sigma^\eta\, M_{\mathcal{Z}}' (S^{-1})'.$$

, and

$$R_{ss} = \sum_{k=-\infty}^{\infty} \operatorname{Cov}\big(H_S(\phi, X_k), H_S(\phi, X_0)\big).$$

Combine both terms to evaluate the integral

$$C = \int_0^\infty e^{nB} R^* e^{nB'}\, dn,$$

, which can be expanded into

$$\begin{aligned} C &= \int_0^\infty \exp\Big(n \begin{bmatrix} D_{\theta\theta} & D_{S\theta} \\ 0 & D_{SS} \end{bmatrix}\Big) \begin{bmatrix} R_{\theta\theta} & 0 \\ 0 & R_{SS} \end{bmatrix} \exp\Big(n \begin{bmatrix} D'_{\theta\theta} & 0 \\ D'_{S\theta} & D'_{SS} \end{bmatrix}\Big)\, dn \\ &= \int_0^\infty \left(\sum_{k=0}^{\infty} \frac{n^k}{k!} \begin{bmatrix} D_{\theta\theta} & D_{S\theta} \\ 0 & D_{SS} \end{bmatrix}^k \right) \begin{bmatrix} R_{\theta\theta} & 0 \\ 0 & R_{SS} \end{bmatrix} \left(\sum_{k=0}^{\infty} \frac{n^k}{k!} \begin{bmatrix} D'_{\theta\theta} & 0 \\ D'_{S\theta} & D'_{SS} \end{bmatrix}^k \right) dn. \end{aligned}$$

Define the off diagonal block of the matrix exponential as

$$\begin{aligned} e^{nD_{S\theta}} &= e^{nD_{\theta\theta}} \int_0^n e^{-uD_{\theta\theta}}\, D_{S\theta}\, e^{uD_{SS}}\, du \\ &= \int_0^n e^{(n-u)D_{\theta\theta}}\, D_{S\theta}\, e^{uD_{SS}}\, du \equiv \Upsilon_{S\theta}(n), \end{aligned}$$

so that we express C as

$$C = \int_0^\infty \begin{bmatrix} e^{nD_{\theta\theta}} & \Upsilon_{S\theta}(n) \\ 0 & e^{nD_{SS}} \end{bmatrix} \begin{bmatrix} R_{\theta\theta} & 0 \\ 0 & R_{SS} \end{bmatrix} \begin{bmatrix} e^{nD'_{\theta\theta}} & 0 \\ \Upsilon'_{S\theta}(n) & e^{nD'_{SS}} \end{bmatrix} dn$$

$$= \int_0^\infty \begin{bmatrix} e^{nD_{\theta\theta}} R_{\theta\theta} & \Upsilon_{S\theta}(n) R_{SS} \\ 0 & e^{nD_{SS}} R_{SS} \end{bmatrix} \begin{bmatrix} e^{nD'_{\theta\theta}} & 0 \\ \Upsilon'_{S\theta}(n) & e^{nD'_{SS}} \end{bmatrix} dn$$

$$= \int_0^\infty \begin{bmatrix} e^{nD_{\theta\theta}} R_{\theta\theta} e^{nD'_{\theta\theta}} + \Upsilon_{S\theta}(n) R_{SS} \Upsilon'_{S\theta}(n) & \Upsilon_{S\theta}(n) R_{SS} e^{nD'_{SS}} \\ e^{nD_{SS}} R_{SS} \Upsilon'_{S\theta}(n) & e^{nD_{SS}} R_{SS} e^{nD'_{SS}} \end{bmatrix} dn.$$

Compare to the result stated in Evans and Honkapohja (2001), the additional covariance term contributes the larger variances in θ.

A.6 E-stabbility Conditions in Extension

In this subsection, we provide the details on the necessary conditions for E-stabile equilibrium in the extension model.

Suppose that public signals are unbiased, that is $\mathbb{E}[G^{CB}] = \Theta^*$ (so $\mathbb{E}[G^{CB}_V] = \Theta^{V*}$, $\mathbb{E}[G^{CB}_X] = \Theta^{X*}$). And, using regularity conditions that $M_Z < \infty$, $E[Z_t \eta_t] = 0$ and Lemma 3.1, the system (27) and (28) can be approximated using stochastic recursive algorithm with the ODE:

$$T^V(\tilde{\Theta}^V_{t|t-1}) = \Omega_1 \tilde{\Theta}^V_{t|t-1} + \Omega_2 (\tilde{\Theta}^V_{t|t-1} \tilde{\Theta}^V_{t|t-1}) + \Omega_3 \tag{47}$$

$$T^X(\tilde{\Theta}^V_{t|t-1}, \tilde{\Theta}^X_{t|t-1}) = \Omega_1 \tilde{\Theta}^X_{t|t-1} + \Omega_2 (\tilde{\Theta}^V_{t|t-1} \tilde{\Theta}^X_{t|t-1} + \tilde{\Theta}^X_{t|t-1} H) + \Omega_4 H. \tag{48}$$

Jacobian for the Θ^V-block. Using $d(AB) = (dA)B + A(dB)$ and $d(A^2) = (dA)A + A(dA)$, and noting $d\tilde{\Theta}^V = \bar{G}_1\, d\Theta^V$, we have

$$\begin{aligned} dT^V &= \Omega_1\, d\tilde{\Theta}^V + \Omega_2 \big((d\tilde{\Theta}^V) \tilde{\Theta}^V + \tilde{\Theta}^V (d\tilde{\Theta}^V) \big) \\ &= \Omega_1 \bar{G}_1\, d\Theta^V + \Omega_2 \bar{G}_1^2 \Big(d\Theta^V (\Theta^V + \rho G^{CB}_V) + (\Theta^V + \rho G^{CB}_V) d\Theta^V \Big). \end{aligned} \tag{49}$$

Vectorizing with $\mathrm{vec}(ABC) = (C' \otimes A)\mathrm{vec}(B)$ gives

$$\mathrm{vec}(dT^V) = \Big[(\Theta^V + \rho G^{CB}_V)' \otimes (\Omega_2 \bar{G}_1^2) + I_3 \otimes \big(\Omega_1 \bar{G}_1 + \Omega_2 \bar{G}_1^2 (\Theta^V + \rho G^{CB}_V) \big) \Big] \mathrm{vec}(d\Theta^V). \tag{50}$$

Therefore,

$$DT_{VV} = (\Theta^V + \rho G^{CB}_V)' \otimes (\Omega_2 \bar{G}_1^2) + I_3 \otimes \big(\Omega_1 \bar{G}_1 + \Omega_2 \bar{G}_1^2 (\Theta^V + \rho G^{CB}_V) \big). \tag{51}$$

Jacobian for the Θ^X-block (conditional on Θ^V). Holding Θ^V fixed (as in the triangular E-stability logic), $d\tilde{\Theta}^V = 0$ and $d\tilde{\Theta}^X = \bar{G}_1\, d\Theta^X$. Then

$$\begin{aligned} dT^X &= \Omega_1\, d\tilde{\Theta}^X + \Omega_2\big(\tilde{\Theta}^V\, d\tilde{\Theta}^X + d\tilde{\Theta}^X\, H\big) \\ &= \Big(\Omega_1 \bar{G}_1 + \Omega_2 \bar{G}_1\, \tilde{\Theta}^V\Big) d\Theta^X + \Omega_2 \bar{G}_1\, d\Theta^X H, \end{aligned} \tag{52}$$

where $\tilde{\Theta}^V = \bar{G}_1(\Theta^V + \rho G_V^{CB})$. Vectorizing yields

$$\operatorname{vec}(dT^X) = \Big[I_3 \otimes \big(\Omega_1 \bar{G}_1 + \Omega_2 \bar{G}_1^2(\Theta^V + \rho G_V^{CB})\big) + H' \otimes (\Omega_2 \bar{G}_1)\Big] \operatorname{vec}(d\Theta^X), \tag{53}$$

so

$$DT_{XX} = I_3 \otimes \big(\Omega_1 \bar{G}_1 + \Omega_2 \bar{G}_1^2(\Theta^V + \rho G_V^{CB})\big) + H' \otimes (\Omega_2 \bar{G}_1). \tag{54}$$

E-stability criterion (block-triangular logic). Because T^V depends only on Θ^V while T^X depends on (Θ^V, Θ^X), the Jacobian in the ordering $(\operatorname{vec}\Theta^V, \operatorname{vec}\Theta^X)$ is block lower-triangular:

$$DT(\Theta) = \begin{bmatrix} DT_{VV} & 0 \\ DT_{XV} & DT_{XX} \end{bmatrix}.$$

Hence the eigenvalues of $DT(\Theta)$ are the union of the eigenvalues of DT_{VV} and DT_{XX}; the off-diagonal block DT_{XV} does not affect the spectrum. Therefore the (local) E-stability condition at the fixed point Θ^* is:

$$\Re\big(\lambda_i(DT_{VV}(\Theta^*))\big) < 1 \quad \text{and} \quad \Re\big(\lambda_i(DT_{XX}(\Theta^*))\big) < 1, \tag{55}$$

with DT_{VV} and DT_{XX} given in (51)–(54).

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